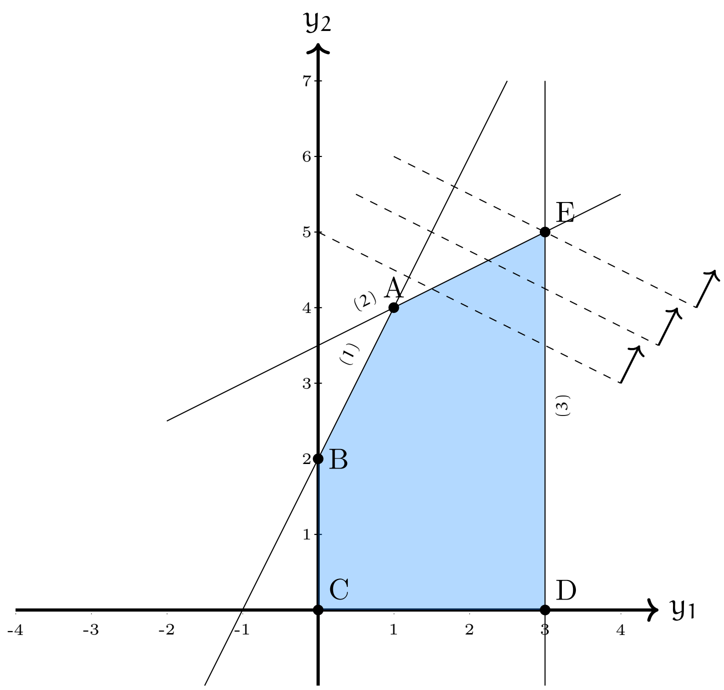
1. The dual problem is

$$\max_{y \in \mathbb{R}^2} \quad y_1 + 2y_2$$

subject to

$$\begin{aligned} -2y_1 + y_2 &\leq 2 \qquad (x_1), \\ -y_1 + 2y_2 &\leq 7 \qquad (x_2), \\ y_1 &\leq 3 \qquad (x_3), \\ y_1, y_2 &\geq 0. \end{aligned}$$



The graphical solution to the dual problem is given in the above figure. The optimal solution to the dual denoted by point E is

$$\begin{pmatrix} y_1^* \\ y_2^* \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix},$$

and the optimal objective function value is 13.

2. The complementarity slackness conditions are stated as follows. Consider the primal linear problem

$$\min_{x \in \mathbb{R}^n} c^T x$$

subject to

$$Ax = b,$$

$$x \geq 0,$$

where $A \in \mathbb{R}^{mxn}$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}^n$, and the dual problem

$$\max_{\lambda \in \mathbb{R}^m} \lambda^T b$$

subject to

$$A^T\lambda \le c.$$

Consider x^* primal feasible and λ^* dual feasible. x^* is optimal for the primal and λ^* is optimal for the dual if and only if

$$(c_i - \sum_{j=1}^{m} a_{ji}\lambda_j^*)x_i^* = 0, \quad i = 1, ..., n.$$

These conditions mean that if a primal variable is non zero, then the corresponding dual constraint must be active, and if the dual constraint is inactive, then the primal variable must be equal to 0.

In order to utilize this theorem, we first write the standard form of the original problem as follows:

$$\min_{x\in\mathbb{R}^5} \quad 2x_1 + 7x_2 + 3x_3$$

subject to

$$\begin{aligned} -2x_1 - x_2 + x_3 - x_4 = 1 &\quad (y_1), \\ x_1 + 2x_2 - x_5 = 2 &\quad (y_2), \\ x_1, x_2, x_3, x_4, x_5 \ge 0. \end{aligned}$$

The dual of the standard form is then

$$\max_{y\in\mathbb{R}^2} \quad y_1 + 2y_2$$

subject to

$$\begin{aligned} -2y_1 + y_2 \le 2 &\quad (x_1), \\ -y_1 + 2y_2 \le 7 &\quad (x_2), \\ y_1 \le 3 &\quad (x_3), \\ -y_1 \le 0 &\quad (x_4), \\ -y_2 \le 0 &\quad (x_5), \\ y_1, y_2 \in \mathbb{R}. \end{aligned}$$

As we have solved the dual problem, we know that

$$\begin{pmatrix} y_1^* \\ y_2^* \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix},$$

and the optimal objective function value is 13. We then write the complementarity slackness conditions that need to be satisfied at optimality:

$$\begin{aligned} (c_1 - (-2y_1^* + y_2^*))x_1^* = 0, \\ (c_2 - (-y_1^* + 2y_2^*))x_2^* = 0, \\ (c_3 - (y_1^*))x_3^* = 0, \\ (c_4 - (-y_1^*))x_4^* = 0, \\ (c_5 - (-y_2^*))x_5^* = 0, \end{aligned}$$

where $c^T = (2, 7, 3, 0, 0)$ and $y_1^* = 3$, $y_2^* = 5$. Using this information, we get the following equalities:

$$\begin{aligned} 3x_1^* &= 0, \\ 0x_2^* &= 0, \\ 0x_3^* &= 0, \\ 3x_4^* &= 0, \\ 5x_5^* &= 0. \end{aligned}$$

We deduce that $x_1^* = x_4^* = x_5^* = 0$. As x^* must be feasible, we replace x_1^*, x_4^*, and x_5^* in the constraints of the original problem:

$$\begin{aligned} -x_2^* + x_3^* &= 1, \\ 2x_2^* &= 2. \end{aligned}$$

Therefore, $x_2^* = 1$ and $x_3^* = 2$. The optimal solution to the primal problem is therefore $(x_1^*, x_2^*, x_3^*, x_4^*, x_5^*) = (0, 1, 2, 0, 0)$ and the objective function value is 13.

3. The primal problem has x_2 and x_3 in the basis at optimality, so that

$$c_B = \begin{pmatrix} 7 \\ 3 \end{pmatrix}.$$

The optimal basis matrix is given by

$$B = \begin{pmatrix} -1 & 1 \\ 2 & 0 \end{pmatrix}.$$

Then,

$$B^T = \begin{pmatrix} -1 & 2 \\ 1 & 0 \end{pmatrix}.$$

The result is then verified as:

$$B^T y^* = B^T \begin{pmatrix} y_1^* \\ y_2^* \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} = c_B.$$